

# Effects of different plasma substitutes on blood coagulation: A comparative review

Evert de Jonge, MD, PhD; Marcel Levi, MD, PhD

**Objective:** To compare the effects of different colloid plasma substitutes on blood coagulation and postoperative blood loss.

**Data Sources:** Relevant studies were obtained from the medical literature.

**Study Selection:** Articles were selected that provided data on the effects of colloids on hemostasis and postoperative blood loss in humans. Studies comparing different colloids were looked for using MEDLINE and by searching through the references of studies as they were collected.

**Data Synthesis:** Articles were reviewed and relevant data were extracted and partly presented in comparative tables.

**Conclusions:** Dextran, gelatin, and hydroxyethyl starch (HES) all can induce a specific decrease of von Willebrand factor and factor VIII:c. Blood coagulation is most impaired by dextran and high molecular weight HES, both associated with increased postoperative blood loss. The effects of HES on blood coagulation have been shown to depend on its molecular weight and rate of

elimination. Detrimental effects have been shown for high molecular weight HES. Medium molecular weight (MMW)-HES with a high degree of substitution (HES 200/0.62) and MMW-HES with high C2/C6 hydroxyethylation ratio (HES 200/0.5/13) are slowly degradable and have been shown to impair blood coagulation after repeated administration. Rapidly degradable HES 200/0.5/6 and gelatin-based plasma expanders appear not to impair hemostasis. However, based on the reviewed literature, all artificial colloids could potentially induce increased bleeding tendency after infusion of very large volumes and especially when given to patients with even mild forms of von Willebrand disease. In those circumstances, crystalloid solutions or alternatives such as plasma or albumin, although associated with other serious complications, could be considered. (Crit Care Med 2001; 29:1261–1267)

**KEY WORDS:** hetastarch; gelatin; albumins; plasma substitutes; dextrans; colloids; blood coagulation; blood coagulation disorders; hemorrhage; surgical blood loss; hemostasis

Synthetic colloids are increasingly used as plasma substitutes in hypovolemic patients because they are readily available, carry no risk of transmitting viral or other plasma transfusion-related disease, and are relatively inexpensive. Controversy exists over whether colloids or crystalloid solutions should be used as plasma expander. Advocates of colloids argue that hypooncotic crystalloids leak from the plasma to excessively expand the interstitial fluid volume, whereas crystalloid supporters argue that leakage of colloid into the interstitial space contributes to edema formation (1). A systematic review of randomized trials of resuscitation with colloids compared with crystalloids

found no difference between these treatments with respect to mortality, pulmonary edema, or length of stay (2).

Considerable differences exist between countries regarding the use of plasma substitutes. In European countries, both gelatins and medium molecular weight starches are used as volume expanders. In the United States on the other hand, high molecular weight starches are most often used, and gelatin-based plasma substitutes are not available at all (3).

Because colloids are frequently used in bleeding patients or in situations with a high risk of bleeding such as trauma or during surgery, there is some concern regarding the effects of colloids on blood coagulation and platelet function. In fact, all three distinct classes of artificial colloids (i.e., dextrans, hydroxyethyl starches, and gelatins) have been associated with derangements of the hemostatic system, although the clinical significance of these derangements is a matter of debate (4). Most evidence points to an impairment of coagulation by plasma substitutes, although some authors, referring to thromboelastography

studies, have suggested that hemodilution per se results in a hypercoagulable state (5, 6). These findings, however, have been disputed (7). In this review we will focus on the antihemostatic effects of artificial colloids and albumin on platelet function and blood coagulation.

## Dextrans

Dextrans are polydisperse glucose polymers produced by bacteria growing in sucrose-containing media. Commercially available dextran-based plasma substitutes have an average molecular weight of 40–70 kDa. Besides their plasma expanding properties, they also exert an anticoagulant effect. Indeed, dextrans have been shown to be effective in preventing postoperative venous thrombosis and pulmonary embolism (8–10). Dextran infusion induces a “von Willebrand syndrome,” with decreased levels of von Willebrand factor (vWF) and associated factor VIII (VIII:c) (11–13). The fall in vWF and factor VIII:c after dextran administration is more than can be explained by its dilutional effects. vWF is the ligand between the platelet surface

From the Departments of Intensive Care (ED) and Vascular Medicine (ML), Academic Medical Center, University of Amsterdam, The Netherlands.

Address requests for reprints to: E. de Jonge, MD, PhD, Department of Intensive Care, Academic Medical Center, P.O. Box 22660, 1100 DD Amsterdam, The Netherlands. E-mail: E.dejonge@amc.uva.nl

Marcel Levi is an investigator of the Royal Netherlands Academy of Science.

Copyright © 2001 by Lippincott Williams & Wilkins

receptor protein GPIb and subendothelial collagen, thereby causing platelet adhesion to the vessel wall. Decreased levels of vWF, therefore, may lead to an impaired primary hemostasis. Indeed, prolonged bleeding times were observed after infusion of dextran in animals (14, 15) as well as in humans (16, 17). The prolongation of bleeding time was totally reversed after increasing vWF levels by the intravenous administration of desmopressin (1-desamino-8-D-arginine vasopressin) (17). Besides their effects on the factor VIII/vWF complex, dextrans also enhance fibrinolysis (18–20). Earlier reports have suggested that fibrinolysis is increased by formation of complexes that include dextran, fibrin, plasminogen activator, and  $\alpha$ -2 antiplasmin. Within these complexes, dextran seems to protect plasmin from the inhibitory effects of  $\alpha$ -2 antiplasmin. Indeed, clots formed in the presence of dextran are relatively bulky in size, exhibit less tensile strength, and seem more easily disrupted (21–23). Recently, it has been shown that fibrinolysis may be enhanced after dextran infusion by increased plasma concentrations of tissue type plasminogen activator (t-PA) and decreased concentrations of the physiologic inhibitor of fibrinolysis plasminogen activator inhibitor-1 (PAI-1) (20).

After its administration, the anti-hemostatic effects of dextran on factor VIII/vWF and fibrinolysis may result in an increased bleeding tendency. In clinical studies comparing dextran with unfractionated heparin, low molecular weight heparin, or the heparinoid Orgaran as prophylaxis against venous thromboembolism, dextran infusion resulted in increased postoperative blood loss after transurethral prostatectomy (24) or orthopedic surgery (25, 26) and increased the need for blood transfusion after surgery for hip fracture (27). In a study comparing dextran with 4% albumin solution as an intraoperative infusion fluid in orthopedic surgical patients who were also treated with low molecular weight heparin thrombosis prophylaxis, patients treated with dextran needed more blood transfusions (28). There are no clinical studies comparing dextran with other synthetic colloid solutions with respect to bleeding complications.

## Gelatins

Gelatins are polydisperse polypeptides produced by degradation of bovine collagen. Two separate forms are produced:

succinylated (modified) gelatins, which have NH<sub>3</sub> groups replaced by COO<sup>−</sup> groups due to reaction of the basic peptide with succinic acid anhydride, and polygelines, consisting of polypeptides cross-linked by urea bonds. Although for a long time gelatins were considered not to influence blood coagulation other than by dilution (29), there is now increasing evidence that gelatins do influence platelet function and blood coagulation. *In vitro* blood coagulation in the presence of gelatin was studied using thromboelastography and scanning electron microscopy. Clots produced in the presence of gelatin had decreased weight and strength and loss of the normal reticular network of fibrin strands (30). Although these results could not be confirmed in another study using thromboelastography, this study also showed a decrease in clot strength after dilution by gelatin as compared with saline dilution (31). Others have studied the effects of *in vitro* dilution of blood on platelet aggregation. It was found that gelatin impaired aggregation induced by ristocetin and polybrene (32). Agglutination tests induced by ristocetin and polybrene are specific for binding of vWF to the platelet receptor GPIb. These findings are supported by the observation that *in vivo* administration of 1 L of a gelatin-based plasma expander to healthy humans induced a von Willebrand like syndrome with lengthening of bleeding time, impaired ristocetin-induced platelet aggregation, and decreased levels of plasma vWF. It was suggested that vWF binds to gelatin by means of its collagen binding site. Also, a decrease in thrombin generation, as measured by thrombin-antithrombin complexes and prothrombin fragment F1 + 2, was observed after gelatin administration, probably caused by hemodilution (33). The clinical relevance of the impairment of hemostasis after gelatin infusion is uncertain. Only one report suggests increased blood loss after cardiac surgery when gelatin instead of albumin was given preoperatively (32). However, other studies comparing gelatin with HES (34) or HES and albumin (35) found no difference or, in some cases, even an improvement (36) in postoperative blood loss when gelatin was given.

## Hydroxyethyl Starch

Hydroxyethyl starch (HES) is generally considered as an effective and safe plasma substitute. However, bleeding

complications have been reported after administration of HES in various clinical settings. High molecular weight HES (HMW-HES, Hetastarch, average molecular weight 480,000 Da), which is the only HES solution that is approved in the United States as a plasma expander, has been associated with increased postoperative bleeding after neurosurgery (37) and in patients undergoing cardiac surgery (35, 38, 39). Furthermore, many case reports have been published describing bleeding complications after HMW-HES in various clinical situations (40–44). It has been suggested that not all HES solutions have negative effects on blood coagulation, but that these effects depend on the average molecular weight of the HES molecules and its kinetics of elimination.

## Pharmacokinetics of HES

Natural starches cannot be used as plasma substitutes because they are rapidly degraded by circulating amylase and they are insoluble at neutral pH. HES are polymers of glucose units derived from amylopectin and modified by substituting hydroxyethyl for hydroxyl groups on glucose molecules. The substitution results in slower degradation and highly increased solubility. HES are very polydisperse solutions of molecules with a broad range of molecular weights from very small to several hundred thousand Dalton. After administration of HES, the low molecular weight fraction is rapidly lost by renal elimination and the large molecules are progressively hydrolyzed, resulting in an average *in vivo* molecular weight that is significantly lower than the average molecular weight of the infused fluid. The rate at which degradation of HES molecules occurs depends on the degree of substitution (DS), that is the proportion of glucose units having a hydroxyethyl group substituted for a hydroxyl group. There are three possible sites for substitution, leading to a maximum DS of three. Currently available HES solutions have a DS of 0.5–0.7. The rate of degradation and elimination is highest with low values for DS (4). Because substitution is possible at positions 2, 3, or 6 of the glucose unit, different patterns of substitution are possible. The substitution pattern is characterized by the C2/C6 hydroxyethylation ratio. Clearance of HES is slowest with high C2/C6 ratios (45). The characteristics of HES solutions are given by their initial molec-

ular weight, the degree of substitution, and the C2/C6 ratio. Thus, HES 200/0.5/6 has an initial average molecular weight of 200,000 Da, with 50% of glucose units having a hydroxyethyl group and with a C2/C6 ratio of 6 (i.e., six times more substitutions at the C2 position as compared with the C6 position).

### Effects on Factor VIII and von Willebrand Factor

Studies in healthy human volunteers have shown that circulating levels of factor VIII and vWF decreased significantly after infusion of 0.5–1 L of HMW-HES (46) or medium molecular weight HES (MMW-HES) (47, 48). Similar reductions in factor VIII and vWF have not only been found in healthy volunteers, but also in a number of clinical studies (40, 49–54). Considerable insight into the influence of HES on blood coagulation comes from observations by Treib et al (45, 55–57). In several 10-day hemodilution experiments in patients, these authors found that a decrease in vWF and factor VIII was only observed when MMW-HES was given that was slowly degradable (i.e., with high degree of substitution and/or high C2/C6 ratio). Those infusions resulted in accumulation of HES molecules with higher molecular weights. It was concluded that the negative effects on hemostasis depended on the *in vivo* molecular weight (45, 55, 56), and that therapy with low-molecular weight HES 70/0.5/4 or easily degradable MMW-HES 200/0.5/6 did not influence blood coagulation (57). However, we recently observed a 33% decrease in vWF and a 28% decrease in factor VIII after administration of 1 L of rapidly degradable HES 200/0.5/6 to healthy volunteers. These decreases were more than could be explained by HES-induced plasma dilution. In accordance with the diminished vWF levels, platelet adhesive function, measured by the platelet function analyzer PFA-100 (Dade Behring, Marburg, Germany), was significantly prolonged after HES as compared with albumin 4%, which is compatible with an acquired von Willebrand's syndrome (E. de Jonge et al., manuscript submitted). A possible explanation for these different findings in healthy volunteers as compared with the studies in patients could be that vWF is an acute phase protein, increasing during acute illness and thereby potentially masking a concomitant HES-induced decrease. Furthermore, relatively low quantities of

HES were given during the hemodilution experiments by Treib et al. (45, 55, 56): (1000 or 500 mL/day). In circumstances when large volumes of HES are given over a short time period (e.g., in bleeding patients with circulatory shock), HES could potentially induce a clinically relevant coagulation defect. Also, uncertainty exists about the clinical effect of the administration of HES on blood coagulation in patients with already low circulating levels of vWF.

### Other Markers of Coagulation and Fibrinolysis

Reductions in the concentrations of other plasma coagulation factors, which could be fully ascribed to plasma dilution after administration of HES, have been reported repeatedly (53, 56, 58, 59). The prothrombin time is only slightly prolonged after the administration of HES, probably due to dilution of plasma factors (5, 46). The effects of HES on the activated partial thromboplastin time (aPTT) depend on its molecular weight and the kinetics of elimination. Significant prolongation of the aPTT up to 40% has been described after repeated infusion of slowly degradable MMW-HES 200/0.62/10. In contrast, only a minimal prolongation was found after the infusion of LMW-HES 70/0.5/4 or easily degradable MMW-HES 200/0.5/6 (45, 48). A significant prolongation of the aPTT has also been found after HMW-HES 480 (40, 46). The effects of HES on the aPTT can be readily explained by the specific decreases in factor VIII, potentially in combination with some dilution of other plasma factors. HMW-HES 480 has been associated with increased fibrinolysis. Hetastarch, but not MMW-HES, induced a significant decrease of urokinase-activated clot lysis time. However, fibrin monomers and fibrin degradation products were not increased (58, 60). Infusion of 500 mL MMW-HES 200/0.5 in healthy volunteers did not result in changes in plasma levels of t-PA, urokinase-plasminogen activator (u-PA), plasminogen activator inhibitor (PAI), plasmin-antiplasmin complexes (PAPc), or D-dimer when compared with infusion of albumin 5%. The effects of HES on coagulation and fibrinolysis also have been studied using thromboelastography, yielding results that are pointing in the same direction. *In vitro* studies suggested prolonged clot formation time and increases in clot lysis after profound hemodilution with HES (61, 62). Finally,

HES might affect platelet function. Platelet volume decreases after infusion of HES, probably because of a shrinkage of platelets by the increased plasma colloid osmotic pressure (45). It has, however, not yet been established whether the decrease in platelet volume after HES administration affects platelet function and bleeding time. Prolonged bleeding times have been reported after HMW-HES (60), and the infusion of MMW-HES has been associated with normal (60) as well as prolonged bleeding times (5, 36). In contrast, in another study comparing the influence of MMW-HES 200/0.5 and albumin on platelet aggregation, no difference between the two colloids was observed (63).

### Albumin

Albumin is generally considered not to influence blood coagulation and is often used as control fluid in studies evaluating the effects of other colloids. Indeed, we recently found that infusion of 1 L albumin 4% to healthy volunteers resulted in only slightly diminished levels of fibrinogen and factors V, VII, VIII:c, and vWF. These decreases could be completely explained by dilution of plasma factors by the infused fluid (E. de Jonge, manuscript submitted). However, it has been reported that albumin induces an impairment of platelet aggregation and prolongation of bleeding time, observations that could be confirmed in our study (64, 65). The mechanism by which platelet function is impaired is uncertain. We did not find any report on an increased bleeding tendency related to the infusion of albumin.

### Pathogenesis of Colloid-Induced von Willebrand Syndrome

The fact that all three classes of artificial colloids share the same vWF-lowering property may suggest that some common pathophysiologic mechanism exists. This postulated common mechanism has not been fully elucidated yet. No change has been observed in the multimeric pattern of vWF after HES, dextran, or gelatin infusion (13, 33, 66). It has been shown that the lowering effects of these colloids on vWF antigen cannot be reproduced *in vitro* (13, 33, 46). Thus, it has been suggested that colloids bind to vWF leading to accelerated *in vivo* elimination (33, 45). This mechanism has also been observed in patients with monoclo-

Table 1. Overview of studies comparing the effects of different plasma substitutes on laboratory markers of coagulation

|                                                      | Colloids                                                     | No. | Population                                                 | Amount of Colloid                                | Parameters Studied                                                                               | Outcome                                                                                                                                      |
|------------------------------------------------------|--------------------------------------------------------------|-----|------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Studies with HMW-HES                                 |                                                              |     |                                                            |                                                  |                                                                                                  |                                                                                                                                              |
| Claes 1992 (52)                                      | 6% HES 450, 5% albumin                                       | 40  | Brain tumor surgery or hysterectomy                        | 1000 mL                                          | APTT, PT, TT, Fbg, VIII:c, vWF                                                                   | Diminished increase of VIII:c and vWF after HMW-HES                                                                                          |
| Brutocao 1996 (67)                                   | 6% HES 450/0.5, 5% albumin                                   | 38  | Cardiac surgery in children                                | Up to 30 mL/kg                                   | PT, aPTT, TT, Fbg, platelets                                                                     | Increased PT in children who received >20 mL/kg HES                                                                                          |
| Studies with MMW-HES and low degree of substitution  |                                                              |     |                                                            |                                                  |                                                                                                  |                                                                                                                                              |
| Beyer 1997 (34)                                      | 6% HES 200/0.5, 3% modified gelatin                          | 46  | Major orthopaedic hip surgery                              | ~2500 mL within 24 hrs                           | PT, TT, Fbg, platelets                                                                           | No difference                                                                                                                                |
| Boldt 1996 (63)                                      | 10% HES 200/0.5, 20% albumin                                 | 56  | Trauma or surgical patients with sepsis                    | ~5 L HES or 3 L albumin in 5 days                | Platelet aggregation induced by ADP epinephrine, and collagen                                    | No difference                                                                                                                                |
| Boldt 1998 (68)                                      | HES 200/0.5, 20% albumin                                     | 300 | Trauma and sepsis patients                                 | ~5 L HES or 2 L albumin in 5 days                | Platelets, AT III, Fbg, aPTT, PT                                                                 | No difference                                                                                                                                |
| Kapiotis 1994 (48)                                   | HES 200/0.5, 5% albumin                                      | 10  | Healthy volunteers, cross-over study                       | 500 mL                                           | Fbg, aPTT, VIII:c, TATc, D-dimer, t-PA, u-PA, PAI, PAPc                                          | Decreased levels of VIII:c after HES                                                                                                         |
| London 1989 (69)                                     | 10% HES 264/0.45, 5% albumin                                 | 94  | Cardiac surgery                                            | ~1800 mL                                         | PT, aPTT, platelets, bleeding time, Fbg, factor V, VII, VIII:c, IX                               | No difference                                                                                                                                |
| London 1992 (70)                                     | 10% HES 264/0.45, 25% albumin                                | 60  | CABG                                                       | 75 g in priming solution                         | PT, aPTT, platelets, bleeding time, Fbg, Factor VII, VIII, IX                                    | Increased aPTT after HES (only immediately at end of bypass)                                                                                 |
| Rackow 1989 (59)                                     | 10% HES 245/0.45, 5% albumin                                 | 20  | Sepsis                                                     | ~1000 mL                                         | Platelets, aPTT, PT, Fbg, VIII:c, bleeding time                                                  | Decreased levels of VIII:c after MMW-HES                                                                                                     |
| Tigchelaar 1997 (51)                                 | 10% HES 200/0.5, 3% modified gelatin, 5% albumin             | 36  | CABG                                                       | Maximum dose: 3000 mL gelatin and 20 mL/kg HES   | Platelet aggregation induced by ristocetin and ADP, <i>in vitro</i> bleeding time, aPTT, PT, vWF | Decreased levels of vWF after MMW-HES and gelatin                                                                                            |
| Vogt 1996 (71)                                       | HES 200/0.5, 5% albumin                                      | 41  | Total hip replacement                                      | ~2500 mL                                         | Platelets, PT, aPTT, TT, Fbg                                                                     | No difference                                                                                                                                |
| Studies with MMW-HES and high degree of substitution |                                                              |     |                                                            |                                                  |                                                                                                  |                                                                                                                                              |
| Treib 1996 (56)                                      | 10% HES 200/0.5, 6% HES 200/0.62                             | 20  | 10 days hemodilution for cerebral circulatory disturbances | 1000 mL/day on days 1–4, 500 mL/day on days 5–10 | PT, aPTT, TT, Fbg, VIII:c, vWF                                                                   | Progressive prolongation of aPTT and decreased levels of VIII:c and vWF after HES 200/0.62                                                   |
| Studies with MMW-HES and high C2/C6 ratio            |                                                              |     |                                                            |                                                  |                                                                                                  |                                                                                                                                              |
| Treib 1995 (55)                                      | 10% HES 200/0.5/13, 10% HES 200/0.5/6                        | 20  | 10 days hemodilution for cerebral circulatory disturbances | 1000 mL/day on days 1–4, 500 mL/day on days 5–10 | PT, aPTT, TT, Fbg, VIII:c, vWF                                                                   | Increased aPTT and decreased levels of VIII:c and vWF after HES 200/0.5/13                                                                   |
| <i>In vitro</i> studies                              |                                                              |     |                                                            |                                                  |                                                                                                  |                                                                                                                                              |
| Egli 1997 (62)                                       | 6% HES 200/0.5, 4% modified gelatin, 5% albumin, 0.9% saline | 96  | <i>In vitro</i> experiments                                | 30% or 60% dilution                              | Thromboelastography                                                                              | Compromised coagulation parameters and increased clot lysis after HES, gelatin, and albumin. Maximum effect found with HES                   |
| Mardel 1996 (72)                                     | 3.5% polygeline, 4% modified gelatin, 0.9% saline            | 20  | <i>In vitro</i> experiments                                | 15% to 75% dilution                              | Clot weights and elasticity by thromboelastography                                               | Decreased clot weights and elasticity after dilution by polygeline and modified gelatin                                                      |
| Mortier 1997 (61)                                    | 6% HES 200/0.5, 4% modified gelatin, 10% dextran 40          | 11  | <i>In vitro</i> experiments                                | 50% dilution                                     | Thromboelastography                                                                              | Increased clot formation time and decreased maximum amplitude after HES. Dilution with dextran resulted in extremely compromised coagulation |
| Jamnicki 1998 (73)                                   | 6% HES 130/0.4, 6% HES 200/0.5, 0.9% NaCl                    | 80  | <i>In vitro</i> experiments                                | 30% and 60% dilution                             | Thromboelastography                                                                              | Comparable decrease in coagulation parameters and increased clot lysis after both HES solutions                                              |

HMW-HES, high molecular weight hydroxyethyl starch; HES, hydroxyethyl starch; MMW-HES, medium molecular weight hydroxyethyl starch; aPTT, activated partial thromboplastin time; PT, prothrombin time; TT, thromboplastin time; Fbg, fibrinogen; VIII:c, associated factor VIII; vWF, von Willebrand factor; ADP, adenosine diphosphate; AT III, antithrombin III; TATc, thrombin-antithrombin complexes; t-PA, tissue type plasminogen activator; u-PA, urokinase-plasminogen activator; PAI, plasminogen activator inhibitor; PAPc, plasmin-antiplasmin complexes; CABG, coronary artery bypass graft.

Table 2. Overview of prospective randomized clinical trials comparing postoperative blood loss after different plasma substitutes

|                                           | Colloids                                                            | No. | Population                              | Amount of Colloid                                      | Outcome                                                                       |
|-------------------------------------------|---------------------------------------------------------------------|-----|-----------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------|
| Dextran vs. albumin<br>Lisander 1996 (28) | Dextran 70, 5% albumin                                              | 40  | Revision hip arthroplasty               | ~10 mL/kg                                              | Increased blood loss and blood transfusion after dextran                      |
| HES vs. gelatin<br>Boldt 1993 (35)        | HMW-HES 450/0.5,<br>MMW-HES 200/0.5,<br>3.5% gelatin, 5% albumin    | 60  | CABG                                    | 1190–1450 mL                                           | Increased blood loss after HMW-HES. No difference between MMW-HES and gelatin |
| Mortelmans 1995 (36)                      | MMW-HES 200/0.5,<br>3% modified gelatin                             | 42  | Total hip replacement                   | ~2000 mL                                               | Increased blood loss after HES                                                |
| Beyer 1997 (34)                           | 6% MMW-HES 200/0.5,<br>3% modified gelatin                          | 46  | Major orthopedic hip surgery            | ~2500 mL within first 24 hrs                           | No difference in blood loss and transfused packed red cells                   |
| Tigchelaar 1997 (51)                      | 10% MMW-HES 200/0.5,<br>3% modified gelatin,<br>4% albumin          | 36  | CABG                                    | Maximum dose: 3000 mL gelatin and 20 mL/kg MMW-HES     | No difference in blood loss                                                   |
| HES vs. albumin<br>Boldt 1993 (35)        | HMW-HES 450/0.5,<br>MMW-HES 200/0.5,<br>3.5% gelatin,<br>5% albumin | 60  | CABG                                    | 1190–1450 mL                                           | Increased blood loss after HMW-HES. No difference between MMW-HES and gelatin |
| Brutocao 1996 (67)                        | 6% HMW-HES 450/0.5,<br>5% albumin                                   | 38  | Cardiac surgery in children             | Up to 30 mL/kg                                         | No difference in blood loss or number of packed red cells transfused          |
| Vogt 1996 (71)                            | MMW-HES 200/0.5,<br>albumin 5%                                      | 41  | Total hip replacement                   | ~2500 mL                                               | No difference in blood loss or number of packed red cells transfused          |
| London 1989 (69)                          | 10% MMW-HES 264/0.45,<br>5% albumin                                 | 94  | After cardiac surgery                   | ~1800 mL                                               | No difference in blood loss or number of blood transfusions                   |
| London 1992 (70)                          | 10% MMW-HES 264/0.45,<br>25% albumin                                | 60  | Cardiopulmonary bypass priming solution | 75 g                                                   | No difference in blood loss or number of blood transfusions                   |
| Tigchelaar 1997 (51)                      | 10% MMW-HES 200/0.5,<br>3% modified gelatin,<br>4% albumin          | 36  | CABG                                    | Maximum dose: 3000 mL gelatin and 20 mL/kg MMW-HES     | No difference in blood loss                                                   |
| Rosencher 1992 (74)                       | MMW-HES 200/0.62,<br>4% albumin                                     | 16  | Total hip replacement                   | 1500–2000 mL                                           | No difference in blood loss                                                   |
| Gelatin vs. albumin<br>Tabuchi 1995 (32)  | Oxypolygelatin,<br>4% albumin                                       | 60  | CABG                                    | 2000 mL as priming solution for extracorporeal circuit | Increased blood loss after gelatin in subgroup also treated with aprotinin    |
| Wahba 1996 (75)                           | Polygeline, 5% albumin                                              | 20  | Cardiac surgery                         |                                                        | No difference in blood loss                                                   |

HMW-HES, high molecular weight hydroxyethyl starch; MMW-HES, medium molecular weight hydroxyethyl starch; CABG, coronary artery bypass graft; HES, hydroxyethyl starch.

nal gammopathy who have an acquired form of von Willebrand disease and accelerated excretion of vWF-immunoglobulin G paraprotein complex (45).

### Comparison of the Different Colloids with Respect to Blood Coagulation

Studies comparing different colloids in regard to laboratory markers of blood coagulation are summarized in Table 1. Direct comparisons are difficult to make, as all studies differ in the plasma substitutes used, the amount of colloid infused, and

the population studied. Furthermore, some studies looked at effects of short-term fluid therapy (e.g., only during surgery) whereas others considered 10-day hemodilution. Nevertheless, some general conclusions can be drawn. First, the effects of HES on coagulation seem to depend on its molecular weight as well as on the rate of *in vivo* degradation. HMW-HES undoubtedly affects blood coagulation, even if given over a limited time period. Slowly degradable MMW-HES (with a high degree of substitution or high C2/C6 hydroxyethylation) ratio also impairs coagulation after re-

peated administration, probably due to accumulation of macromolecules. In contrast, most studies consider easily degradable MMW-HES (with a low degree of substitution and low C2/C6 ratio) or low molecular weight (LMW)-HES to have minimal influence, if any, on blood coagulation when compared with albumin. The effects of modified gelatin solutions seemed to be similar to the effects of easily degradable MMW-HES and albumin.

An overview of studies comparing different colloids regarding postoperative blood loss is given in Table 2. One single

study found increased postoperative blood loss after dextran administration as compared with albumin. Parallel to the observations on laboratory markers of coagulation, administration of HMW-HES may lead to increased blood loss. No study documented increased bleeding tendency after MMW-HES with low degree of substitution when compared with albumin, suggesting that rapidly degradable MMW-HES can safely be given during surgery. Only one study has been published that compared slowly degradable MMW-HES 200/0.62 with albumin in orthopaedic patients. In this small study, no difference in postoperative blood loss was found. Studies comparing gelatin with albumin or MMW-HES found no difference, except for one study that observed decreased blood loss after gelatin when compared with MMW-HES and one that found increased blood loss when compared with albumin, but only in a subgroup also treated with aprotinin (32, 36). Thus, it seems that dextran and HMW-HES can induce an increased bleeding tendency, whereas rapidly degradable MMW-HES and gelatin are probably safe in this respect. There are, however, two major limitations to this conclusion. First, no studies have addressed the risk of bleeding after repeated administration of colloids. Theoretically, this could easily increase the risk of bleeding, especially with infusion of slowly degradable HES. Second, these conclusions are probably only valid in subjects with normal levels of vWF or even increased levels due to the acute phase response in acutely ill patients. Because all artificial colloids can lead to decreased levels of vWF, they should be given cautiously to patients who are known to have even mild forms of von Willebrand disease. In those patients, especially when they are bleeding, crystalloid solutions or alternatives like plasma or albumin, although associated with other serious complications, could be considered.

## REFERENCES

- Marik PE, Iglesias J: Would the colloid detractors please sit down! *Crit Care Med* 1999; 28:2652-2654
- Choi PT, Yip G, Quinonez LG, et al: Crystalloids vs. colloids in fluid resuscitation: A systematic review. *Crit Care Med* 1999; 27: 200-210
- Treib J, Baron JF, Grauer MT, et al: An international view of hydroxyethyl starches. *Intensive Care Med* 1999; 25:258-268
- Salmon JB, Mythen MG: Pharmacology and physiology of colloids. *Blood Rev* 1993; 7:114-120
- Ruttmann TG, James MF, Aronson I: *In vivo* investigation into the effects of haemodilution with hydroxyethyl starch (200/0.5) and normal saline on coagulation. *Br J Anaesth* 1998; 80:612-616
- Ruttmann TG, James MF, Viljoen JF: Clot formation and gelatin-based plasma substitutes. *Lancet* 1996; 348:1249-1250
- Baddeley DT, Saunders FM, Ollerenshaw L, et al: Clot formation and gelatin-based plasma substitutes. *Lancet* 1996; 347:1763
- Bergqvist D: Modern aspects of prophylaxis and therapy for venous thrombo-embolic disease. *Aust N Z J Surg* 1998; 68:463-468
- Clagett GP, Anderson FAJ, Geerts W, et al: Prevention of venous thromboembolism. *Chest* 1998; 114:531S-560S
- Ljungstrom KG: Dextran prophylaxis of fatal pulmonary embolism. *World J Surg* 1983; 7:767-772
- Aberg M, Hedner U, Bergentz SE: Effect of dextran on factor VIII (antihemophilic factor) and platelet function. *Ann Surg* 1979; 189:243-247
- Aberg M, Hedner U, Bergentz SE: The antithrombotic effect of dextran. *Scand J Haematol Suppl* 1979; 34:61-68
- Battle J, del Rio F, Lopez FM, et al: Effect of dextran on factor VIII/von Willebrand factor structure and function. *Thromb Haemost* 1985; 54:697-699
- Zhang B, Wieslander JB: Dextran's antithrombotic properties in small arteries are not altered by low-molecular-weight heparin or the fibrinolytic inhibitor tranexamic acid: An experimental study. *Microsurgery* 1993; 14:289-295
- Salemark L, Knudsen F, Dougan P: The effect of dextran 40 on patency following severe trauma in small arteries and veins. *Br J Plast Surg* 1995; 48:121-126
- Bergman A, Andreen M, Blomback M: Plasma substitution with 3% dextran-60 in orthopaedic surgery: influence on plasma colloid osmotic pressure, coagulation parameters, immunoglobulins and other plasma constituents. *Acta Anaesthesiol Scand* 1990; 34: 21-29
- Lethagen S, Rugarn P, Aberg M, et al: Effects of desmopressin acetate (DDAVP) and dextran on hemostatic and thromboprophylactic mechanisms. *Acta Chir Scand* 1990; 156: 597-602
- Carlin G, Modig J, Saldeen T: Effect of infusion of dextran 70 on fibrinolysis inhibition activity in human serum. *Acta Chir Scand* 1979; 145:129-131
- Carlin G, Karlstrom G, Modig J, et al: Effect of dextran on fibrinolysis inhibition activity in the blood after major surgery. *Acta Anaesthesiol Scand* 1980; 24:375-378
- Eriksson M, Saldeen T: Effect of dextran on plasma tissue plasminogen activator (t-PA) and plasminogen activator inhibitor-1 (PAI-1) during surgery. *Acta Anaesthesiol Scand* 1995; 39:163-166
- Strauss RG: Volume replacement and coagulation: A comparative review. *J Cardiothorac Anesth* 1988; 2(Suppl 1): 24-32
- Carr ME, Gabriel DA: The effect of dextran 70 on the structure of plasma-derived fibrin gels. *J Lab Clin Med* 1980; 96:985-993
- Ts'ao CH, Krajewski DV: Effect of dextran on platelet activation by polymerizing fibrin. *Am J Pathol* 1982; 106:1-7
- Hjertberg H, Olsson J, Ekstrom T, et al: Do low molecular weight heparin and dextran increase the blood loss in transurethral resection of the prostate? *Br J Urol* 1996; 78: 897-900
- Gruber UF: Prevention of fatal postoperative pulmonary embolism by heparin dihydroergotamine or Dextran 70. *Br J Surg* 1982; 69(Suppl):S54-S58
- Anonymous. Low-molecular-weight heparin (enoxaparin) vs dextran 70: The prevention of postoperative deep vein thrombosis after total hip replacement. The Danish Enoxaparin Study Group. *Arch Intern Med* 1991; 151: 1621-1624
- Bergqvist D, Kettunen K, Fredin H, et al: Thromboprophylaxis in patients with hip fractures: A prospective, randomized, comparative study between Org 10172 and dextran 70. *Surgery* 1991; 109:617-622
- Lisander B, Jacobsson SA, Ivarsson I, et al: Giving both enoxaparin and dextran increases the need for transfusion in revision hip arthroplasty. *Eur J Surg* 1996; 162: 861-866
- Lundsgaard-Hansen P, Tschirren B: Clinical experience with 120,000 units of modified fluid gelatin. *Dev Biol Stand* 1980; 48: 251-256
- Mardel SN, Saunders FM, Allen H, et al: Reduced quality of clot formation with gelatin-based plasma substitutes. *Br J Anaesth* 1998; 80:204-207
- Ruttmann TG, James MF, Viljoen JF: Haemodilution induces a hypercoagulable state. *Br J Anaesth* 1996; 76:412-414
- Tabuchi N, de Haan J, Gallandat Huet RC, et al: Gelatin use impairs platelet adhesion during cardiac surgery. *Thromb Haemost* 1995; 74:1447-1451
- de Jonge E, Levi M, Berends F, et al: Impaired haemostasis by intravenous administration of a gelatin-based plasma expander in human subjects. *Thromb Haemost* 1998; 79: 286-290
- Beyer R, Harmening U, Rittmeyer O, et al: Use of modified fluid gelatin and hydroxyethyl starch for colloidal volume replacement in major orthopaedic surgery. *Br J Anaesth* 1997; 78:44-50
- Boldt J, Knothe C, Zickmann B, et al: Influence of different intravascular volume therapies on platelet function in patients undergoing cardiopulmonary bypass. *Anesth Analg* 1993; 76:1185-1190
- Mortelmans YJ, Vermaut G, Verbruggen AM,

- et al: Effects of 6% hydroxyethyl starch and 3% modified fluid gelatin on intravascular volume and coagulation during intraoperative hemodilution. *Anesth Analg* 1995; 81: 1235-1242
37. Trumble ER, Muizelaar JP, Myseros JS, et al: Coagulopathy with the use of hetastarch in the treatment of vasospasm. *J Neurosurg* 1995; 82:44-47
38. Cope JT, Banks D, Mauney MC, et al: Intraoperative hetastarch infusion impairs hemostasis after cardiac operations. *Ann Thorac Surg* 1997; 63:78-82
39. Villarino ME, Gordon SM, Valdon C, et al: A cluster of severe postoperative bleeding following open heart surgery. *Infect Control Hosp Epidemiol* 1992; 13:282-287
40. Dalrymple-Hay M, Aitchison R, Collins P, et al: Hydroxyethyl starch induced acquired von Willebrand's disease. *Clin Lab Haematol* 1992; 14:209-211
41. Panchal HI, Ramwell J, Lawler PG: Severe coagulopathy in Jehovah's witness. *Anaesthesia* 1989; 44:71-72
42. Damon L, Adams M, Stricker RB, et al: Intracranial bleeding during treatment with hydroxyethyl starch. *N Engl J Med* 1987; 317: 964-965
43. Cully MD, Larson CPJ, Silverberg GD: Hetastarch coagulopathy in a neurosurgical patient. *Anesthesiology* 1987; 66:706-707
44. Lockwood DN, Bullen C, Machin SJ: A severe coagulopathy following volume replacement with hydroxyethyl starch in a Jehovah's Witness. *Anaesthesia* 1988; 43:391-393
45. Treib J, Haass A, Pindur G: Coagulation disorders caused by hydroxyethyl starch. *Thromb Haemost* 1997; 78:974-983
46. Stump DC, Strauss RG, Henriksen RA, et al: Effects of hydroxyethyl starch on blood coagulation, particularly factor VIII. *Transfusion* 1985; 25:349-354
47. Korttila K, Grohn P, Gordin A, et al: Effect of hydroxyethyl starch and dextran on plasma volume and blood hemostasis and coagulation. *J Clin Pharmacol* 1984; 24:273-282
48. Kapiotis S, Quehenberger P, Eichler HG, et al: Effect of hydroxyethyl starch on the activity of blood coagulation and fibrinolysis in healthy volunteers: Comparison with albumin. *Crit Care Med* 1994; 22:606-612
49. Kuitunen A, Hynynen M, Salmenpera M, et al: Hydroxyethyl starch as a prime for cardiopulmonary bypass: effects of two different solutions on haemostasis. *Acta Anaesthesiol Scand* 1993; 37:652-658
50. Halonen P, Linko K, Myllyla G: A study of haemostasis following the use of high doses of hydroxyethyl starch 120 and dextran in major laparotomies. *Acta Anaesthesiol Scand* 1987; 31:320-324
51. Tigchelaar I, Gallandat HR, Korsten J, et al: Hemostatic effects of three colloid plasma substitutes for priming solution in cardiopulmonary bypass. *Eur J Cardiothorac Surg* 1997; 11:626-632
52. Claes Y, Van Hemelrijck J, Van Gerven M, et al: Influence of hydroxyethyl starch on coagulation in patients during the perioperative period. *Anesth Analg* 1992; 75:24-30
53. Conroy JM, Fishman RL, Reeves ST, et al: The effects of desmopressin and 6% hydroxyethyl starch on factor VIII:C. *Anesth Analg* 1996; 83:804-807
54. Lazarchick J, Conroy JM: The effect of 6% hydroxyethyl starch and desmopressin infusion on von Willebrand factor: Ristocetin cofactor activity. *Ann Clin Lab Sci* 1995; 25: 306-309
55. Treib J, Haass A, Pindur G, et al: HES 200/0.5 is not HES 200/0.5. Influence of the C2/C6 hydroxyethylation ratio of hydroxyethyl starch (HES) on hemorheology, coagulation and elimination kinetics. *Thromb Haemost* 1995; 74:1452-1456
56. Treib J, Haass A, Pindur G, et al: All medium starches are not the same: Influence of the degree of hydroxyethyl substitution of hydroxyethyl starch on plasma volume, hemorheologic conditions, and coagulation. *Transfusion* 1996; 36:450-455
57. Stoll M, Treib J, Schenck JF, et al: No coagulation disorders under high-dose volume therapy with low-molecular-weight hydroxyethyl starch. *Haemostasis* 1997; 27:251-258
58. Strauss RG, Stump DC, Henriksen RA, et al: Effects of hydroxyethyl starch on fibrinogen, fibrin clot formation, and fibrinolysis. *Transfusion* 1985; 25:230-234
59. Rackow EC, Mecher C, Astiz ME, et al: Effects of pentastarch and albumin infusion on cardiorespiratory function and coagulation in patients with severe sepsis and systemic hypoperfusion. *Crit Care Med* 1989; 17: 394-398
60. Strauss RG, Stansfield C, Henriksen RA, et al: Pentastarch may cause fewer effects on coagulation than hetastarch. *Transfusion* 1988; 28:257-260
61. Mortier E, Ongenae M, De Baerdemaeker L, et al: In vitro evaluation of the effect of profound haemodilution with hydroxyethyl starch 6%, modified fluid gelatin 4% and dextran 40 10% on coagulation profile measured by thromboelastography. *Anaesthesia* 1997; 52:1061-1064
62. Egli GA, Zollinger A, Seifert B, et al: Effect of progressive haemodilution with hydroxyethyl starch, gelatin and albumin on blood coagulation. *Br J Anaesth* 1997; 78:684-689
63. Boldt J, Muller M, Heesen M, et al: Influence of different volume therapies on platelet function in the critically ill. *Intensive Care Med* 1996; 22:1075-1081
64. Jorgensen KA, Stoffersen E: On the inhibitory effect of albumin on platelet aggregation. *Thromb Res* 1980; 17:13-18
65. Kim SB, Chi HS, Park JS, et al: Effect of increasing serum albumin on plasma D-dimer, von Willebrand factor, and platelet aggregation in CAPD patients. *Am J Kidney Dis* 1999; 33:312-317
66. Treib J, Haass A, Pindur G, et al: Highly substituted hydroxyethyl starch (HES200/0.62) leads to type-I von Willebrand syndrome after repeated administration. *Haemostasis* 1996; 26:210-213
67. Brutocao D, Bratton SL, Thomas JR, et al: Comparison of hetastarch with albumin for postoperative volume expansion in children after cardiopulmonary bypass. *J Cardiothorac Vasc Anesth* 1996; 10: 348-351
68. Boldt J, Muller M, Mentges D, et al: Volume therapy in the critically ill: Is there a difference? *Intensive Care Med* 1998; 24:28-36
69. London MJ, Ho JS, Friedman JK, et al: A randomized clinical trial of 10% pentastarch (low molecular weight hydroxyethyl starch) versus 5% albumin for plasma volume expansion after cardiac operations. *J Thorac Cardiovasc Surg* 1989; 97:785-797
70. London MJ, Franks M, Verrier ED, et al: The safety and efficacy of ten percent pentastarch as a cardiopulmonary bypass priming solution. A randomized clinical trial. *J Thorac Cardiovasc Surg* 1992; 104:284-296
71. Vogt NH, Bothner U, Lerch G, et al: Large-dose administration of 6% hydroxyethyl starch 200/0.5 total hip arthroplasty: Plasma homeostasis, hemostasis, and renal function compared to use of 5% human albumin. *Anesth Analg* 1996; 83:262-268
72. Mardel SN, Saunders F, Ollerenshaw L, et al: Reduced quality of in-vitro clot formation with gelatin-based plasma substitutes. *Lancet* 1996; 347:825
73. Jamnicki M, Zollinger A, Seifert B, et al: Compromised blood coagulation: An *in vitro* comparison of hydroxyethyl starch 130/0.4 and hydroxyethyl starch 200/0.5 using thrombelastography. *Anesth Analg* 1998; 87: 989-993
74. Rosencher N, Vassilief N, Guigonis V, et al: [Comparison of effects of Elohes and albumin on hemostasis in orthopedic surgery]. *Ann F Anesth Reanim* 1992; 11:526-530
75. Wahba A, Sendtner E, Birnbaum DE: Fluid resuscitation with Haemaccel vs. human albumin following coronary artery bypass grafting. *Thorac Cardiovasc Surg* 1996; 44: 178-182